Baochen Tian

Visiting Student, CLEAR Lab, Southern University of Science and Technology
baochen.tian.phys@gmail.com | github.com/bcTiann | Website: bctiann.github.io

EDUCATION

Australian National University Graduated 2025

Bachelor of Science in Physics, 2023 - 2025 | GPA: 6.13 / 7.0
Exchange: National Taiwan University, Spring 2025

Beijing Institute of Technology 2020 - 2023

Undergraduate studies; transferred to ANU in 2023
International Economics and Trade (2020 - 2021); Optical-Electronics Information Science (2021 - 2023)

PERSONAL PROJECTS

SuperMimic Reproduction and Video-to-Robot Adaptation October 2026

Integrated GVHMR and GMR with UniSuperMimic to convert video demonstrations into G1 motion references for policy fine-tuning and MuJoCo evaluation.
Reproduced the 40-motion benchmark on an M4 Pro: 40/40 full clips completed; mean normalized return 0.826 at update 11,000 (single seed; training motions).
Code: github.com/bcTiann/Uni-SuperMimic | Results

GeoRA Reimplementation and GRPO Experiments October 2026

Reimplemented GeoRA for Qwen2.5-1.5B-Instruct and integrated it with GRPO training, mixed precision and checkpoint recovery.
Ran a matched LoRA pilot on all 500 MATH500 questions; separately checked vLLM/TRL policy synchronization.
Code: github.com/bcTiann/geora-repro | Results

Flow Matching Reimplementation of DiffPhyCon 2026

Derived and implemented a flow-matching formulation of the joint trajectory model in DiffPhyCon (NeurIPS 2024), a diffusion-based physical control framework. Matched original model capacity and training configuration; compared 1D Burgers control and 2D Jellyfish state-control trajectories with diffusion baselines.
Investigated guidance and prior reweighting under flow matching using score-to-velocity conversions.
Code: github.com/bcTiann/diffphycon

RESEARCH EXPERIENCE

CLEAR Lab, Southern University of Science and Technology November 2026 - Present

Visiting Student; supervised by Dr. Xiong Zeng in Prof. Wei Zhang's group.

Research on reinforcement learning for dexterous hand control and post-training of embodied foundation models.

Python Pipeline for Post-processing QUOKKA Simulations July 2025 - Present

Research School of Astronomy and Astrophysics, ANU.
Research Assistant (April 2026 - Present). Supervisor: Dr. ChongChong He.

Developing a reusable Python post-processing pipeline for QUOKKA, a next-generation GPU-native astrophysical hydrodynamics code. Covers 3D data loading, chunked computation, analysis and visualisation; produces synthetic images, spectra and statistics for comparisons with astronomical observations and physical model tests.
Using DESPOTIC, Cloudy and analytic line-emission models, with thermal broadening and dust attenuation, to investigate how reliably CO, H-alpha, C+ and H I line widths recover mass-weighted gas velocity dispersion.
Code: github.com/bcTiann/quokka_postprocess_3D

Deep Learning for Cosmic Microwave Background Inpainting February - June 2025

National Taiwan University (exchange). Supervisor: Prof. Pisin Chen; mentor: Reyhan D. Lambaga.

Applied Transformers and DDPMs to inpaint CMB maps; compared angular power spectra with ground truth.

Axion Haloscope Modelling (EQUS Scholarship) January - February 2024

University of Western Australia. Supervisors: Prof. Michael Tobar and Dr. Aaron Quiskamp.

Six-week project: COMSOL modelling of resonant modes and sensitivity for tunable-wall rectangular haloscopes.

SKILLS & INTERESTS

Deep learning: PyTorch, Transformers, diffusion models, flow matching, GRPO and LoRA/GeoRA.
Scientific computing: Python, MuJoCo, QUOKKA post-processing, DESPOTIC and Cloudy.
Chamber music: Co-organised ensembles with friends at BIT and ANU; choir accompanist and public performances.